

ENTREPRENEURSHIP & BUSINESS DESIGN

B. Tech. CSE. III Year, I semester Section D, (9ZC10). Unit - II

**Sreenidhi Institute of Science & Technology
Hyderabad, 501301, India (Sep. 2025)**

Syllabus (Unit -2)

UNIT-II. UNIT II: Introduction to UI / UX: Human centered design and benefits, the distinction between UX and UI, UX process — user research, prototyping strategies, UI principles, UI analysis, UI design, UI components and Responsive design.

What is “Business design”?

Business design is an interdisciplinary field that blends

**Design thinking,
Strategy, and
Business principles**

to create innovative and viable business models.

It's a human-centered approach that uses design methods to solve business problems, focusing on understanding customer needs to build sustainable, market-ready businesses rather than just desirable products.

Human Centered Design and Benefits

- Human-centered design (HCD) is an “iterative process for problem-solving in which the priorities, needs, and behaviors of end-users...serve as a driving force throughout a [solution’s] conceptualization, development, and distribution.”

Human-Centered Design and its Benefits

- **Human-centered design (HCD) is a creative approach to problem-solving.**
- **It starts with the people you are designing for and ends with solutions that are tailored to their needs.**
- **The core idea is to truly understand the users and their problems before you start building a solution.**

Human-Centered Design and its Benefits (..Cont.,)

Illustration: A company wants to create a new banking app. Instead of just building a standard app, they use HCD.

- They interview potential users:
 - To understand their frustrations with current banking apps,
 - Their financial goals, and
 - What they would find helpful.
- They learn that many users are confused by complex jargon.
- Based on this, they design an app with simple language, clear navigation, and a balance budget-tracking feature that users actually want.

Human-Centered Design and its Benefits (..Cont..)

■ Benefits:

Better Products: Products are more likely to be successful because they directly address real user needs.

Increased User Satisfaction: Users are happier and more loyal when a product is easy to use and provides value.

Higher ROI (Return on Investment): Companies save money by avoiding the development of features no one will use and by reducing the need for costly redesigns later on.

Human Centered Design and Benefits

- Human-centered Design, a framework for identifying and designing solutions for the people who use your products.



Human Centered Design and Benefits

- Human-centered Design, a framework for identifying and designing solutions for the people who use your products.



Empathize

**Develop deep
understanding
of the
challenge**

**Clearly articulate
the problem you
want to solve**

Define

**Brainstorm
Potential solutions;
select & deliver
your solution**

Ideate

**Design
prototype (s) to
test your
solution**

Prototype

**Engage innovation
& Improve your
design**

Test

Human Centered Design and Benefits

PHASE I: RESEARCH



WHY START WITH RESEARCH?

You cannot design apart from the world in which your users live.

RESEARCH HELPS YOU LEARN

Who is your target audience?

What are their goals and needs?

What problems can be solved?

Human Centered Design and Benefits

PHASE II: COME UP WITH IDEAS TO SOLVE PROBLEMS



Human Centered Design and Benefits

PHASE III: TEST YOUR ASSUMPTIONS

WHY PROTOTYPE?

You can test your ideas with users

It's faster than coding

It uncovers subtleties & nuances

It's cheaper to fix now than after it's built

PHASE IV: IS WHAT YOU MADE EASY TO USE?

EVALUATION HELPS YOU ANSWER

Is it easy to use?

Does it address the user need?

What works?

What needs improvement?

What's missing?

Human Centered Design and Benefits

PHASE V: LAUNCH 🚀

WHY MEASURE AFTER LAUNCHING?

Product is in the real world

It's being used by your customers

Helps you answer "Did we get it right?"

SOME MEASUREMENT METHODS

Talk to people who are using your product

Look at use analytics

Run surveys

Human Centered Design and Benefits

1. Sales and support reps will be more **accessible to your customers.**
2. Sales and support reps will be able to **customize solutions**, personalize interactions, and **strengthen your relationship** with customers—all of which lead to a **better customer experience.**
3. **You'll better understand** the needs of your customers.
4. You'll increase brand loyalty.
5. Customer **retention will increase.**

Human Centered Design and Benefits

- 7. Your business will gain a competitive edge and be positioned as a leader within your industry.**
- 8. Teams will fix problems at the root rather than just fixing the symptoms.**
- 9. Sales and profit margins will increase.**
- 10. You'll learn more about the products and services what customers want and what they're willing to pay for them.**
- 11. Leveraging automation will help you reduce errors and risks.**

Defining UI, and UX

UI (User Interface):

- The "I" in UI stands for interface.
- It is the visual part of the product.
- UI designers are responsible for the look and feel.
- They choose the colors, fonts, and layout.
- Think of UI as the paint on the house.
- They ask questions like: *Is the button a clear call to action?*
- *Does the color scheme create a cohesive feel?*

Defining UI, and UX (...Cont.,)

UX (User Experience):

- The "X" in UX stands for experience.
- It's the total sum of all the user's interactions with a product.
- Think of it as the blueprint for the entire product.
- UX designers focus on the product's:
 Functionality, Usability, and
 Overall flow.
- They ask questions like:
 Does the product solve a user's problem?
 Is it intuitive?

UI and UX

UI (User Interface):

- This is the specific visual part of the product that a user interacts with
- It includes:
 - Layout,
 - Buttons,
 - Icons,
 - Colors, and typography.
- The UI is what the user sees and clicks on.



The User Interface Design Elements

1. **Input Controls:** AC/DC plug-in socket, USB socket to printer,..
2. **Navigation components:** Menu, Search bars, breadcrumbs, and pagination.
3. **Informational components:** Notifications, tooltips, progress bars, and message boxes.
4. **Containers:** Fetching data from external sources and passing it down to presentational components.

UI and UX (**..cont..**)

In the context of technology and business, UI and UX are two interconnected but distinct fields focused on creating digital products that are both effective and enjoyable for the user.

UX (User Experience):

- This is about how a user feels when they interact with a product, system, or service.**
 - It is the overall experience:**
 - Is it easy to use?**
 - Is it useful?**
 - Is it a pleasure to interact with?**
- UX is a broad field that encompasses everything from market research to content strategy.**

The Distinction Between UI and UX.

Why & how

	UI (User Interface)	UX (User Experience)
Focus	How the product looks.	How the product works and feels.
Goal	To make the product visually appealing and interactive.	To make the product useful, usable, and valuable.
Analogy	The interior design and decorations of the house.	The architecture of a house. Structural integrity (Column & Beams, foundation, F of S)..
Questions	"What color should the 'Buy Now' button be?"	"How does the user navigate from the homepage to the checkout screen?"

Difference between UX - UI

UX

HUMAN-FIRST APPROACH TO PRODUCT DESIGN

- APPLICATION:**
Physical and digital products
- FOCUS:**
The full experience from a user's first contact to the last
- CREATES:**
Structural design solutions for pain points that users encounter anywhere along their journey with the product
- RESULTS IN:**
Products that delight users with their effectiveness

UI

HUMAN-FIRST APPROACH TO DESIGNING THE AESTHETIC EXPERIENCE OF A PRODUCT

- APPLICATION:**
Digital products only
- FOCUS:**
Visual touchpoints that allow users to interact with a product
- CREATES:**
Combinations of typography, color palettes, buttons, animations, and imagery
- RESULTS IN:**
Products that delight users aesthetically

UX Design

- **UX design refers to the term “user experience design”, while UI stands for “user interface design”. Both elements are crucial to a product and work closely together.**
- **But despite their professional relationship, the roles themselves are quite different, referring to very different aspects of the product development process and the design discipline.**

User Experience (UX) Design

- User experience design is a human-first way of designing products. Don Norman, a cognitive scientist and co-founder of the **Nielsen Norman Group Design Consultancy**, is credited with coining the term “user experience” in the late 1990s. Here’s how he describes it:
- “User experience encompasses all aspects of the end-user’s interaction with the company, its services, and its products.”

- **What does UX design involve?**
- A UX designer thinks about how the experience makes the user feel, and how easy it is for the user to accomplish their desired tasks. They also observe and conduct **task analyses** to see how users actually complete tasks in a user flow.
- For example: How easy is the checkout process when shopping online? How easy is it for you to grip that vegetable peeler? Does your online banking app make it easy for you to manage your money?
- The ultimate purpose of UX design is to create easy, efficient, relevant, and all-round pleasant experiences for the user.

- User experience design is the process of developing and improving the quality of interaction between a user and all facets of a company.
- User experience design is, in theory, a non-digital (cognitive science) practice, but used and defined predominantly by digital industries.
- UX design is NOT about visuals; it focuses on the overall feel of the experience.

What is user interface (UI) design?

- While user experience is a conglomeration of tasks focused on the optimization of a product for effective and enjoyable use, user interface design is its complement; the look and feel, the presentation and interactivity of a product.

- A user interface is the point of interaction between the user and a digital device or product—like the touch screen on your smart phone, or the touchpad you use to select what kind of coffee you want from the coffee machine.
- In relation to websites and apps, UI design considers the look, feel, and interactivity of the product. It's all about making sure that the user interface of a product is as intuitive as possible, and that means carefully considering each and every visual, interactive element the user might encounter.

UI Design Involve

- ❖ **User interface design is a purely digital practice. It considers all the visual, interactive elements of a product interface—including buttons, icons, spacing, typography, color schemes, and responsive design.**
- ❖ **The goal of UI design is to visually guide the user through a product's interface. It's all about creating an intuitive experience that doesn't require the user to think too much!**
- ❖ **UI design transfers the brand's strengths and visual assets to a product's interface, making sure the design is consistent, coherent, and aesthetically pleasing.**



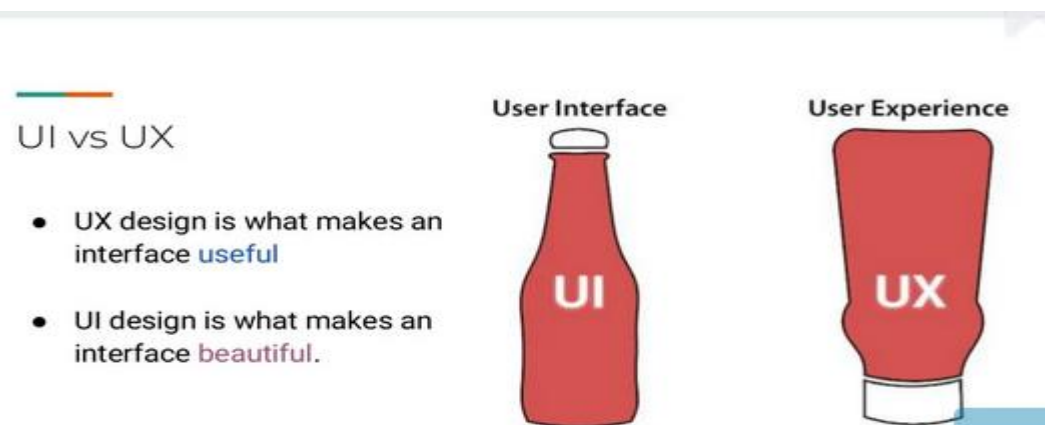
**Poor flow
User time is wasted.
Frustrated customer.**



**Quick flow because of wider
lid & always rests on lid.
User time is respected.
Happier customer.**

Difference between UX and UI design

- If you imagine a product as the human body, the bones represent the code which give it structure. The organs represent the UX design: measuring and optimizing against input for supporting life functions. UI design represents the cosmetics of the body; its presentation, its senses and reactions.
- UI is the saddle, the stirrups, and the reins. UX is the feeling you get being able to ride the horse.” —Dain Miller, Web Developer



- **The main difference to bear in mind is this:** UX design is all about the overall feel of the experience, while UI design is all about how the product's interfaces look and function.
- **A UX designer considers the user's entire journey** to solve a particular problem; what steps do they take? What tasks do they need to complete? How straightforward is the experience?
- Much of their work focuses on finding out what kinds of problems and pain-points users come up against, and how a certain product might solve them. They'll conduct extensive user research in order to find out who the target users are and what their needs are in relation to a certain product.

- With the skeleton of the product mapped out, the UI designer steps in to bring it to life. The UI designer considers all the visual aspects of the user's journey, including all the individual screens and touch points that the user might encounter; think tapping a button, scrolling down a page or swiping through an image gallery.



- UX design is all about identifying and solving user problems; UI design is all about creating intuitive, aesthetically-pleasing, interactive interfaces.
- UX design usually comes first in the product development process, followed by UI. The UX designer maps out the bare bones of the user journey; the UI designer then fills it in with visual and interactive elements.
- UX can apply to any kind of product, service, or experience; UI is specific to digital products and experiences.

Responsiveness and Interactivity

- UI prototyping
- Interactivity and animation
- Adaptation to all device screen sizes
- Implementation with developer

Prototyping and its remarkable reasons



3 Essentials for successful prototyping



**Speed is
Everything**

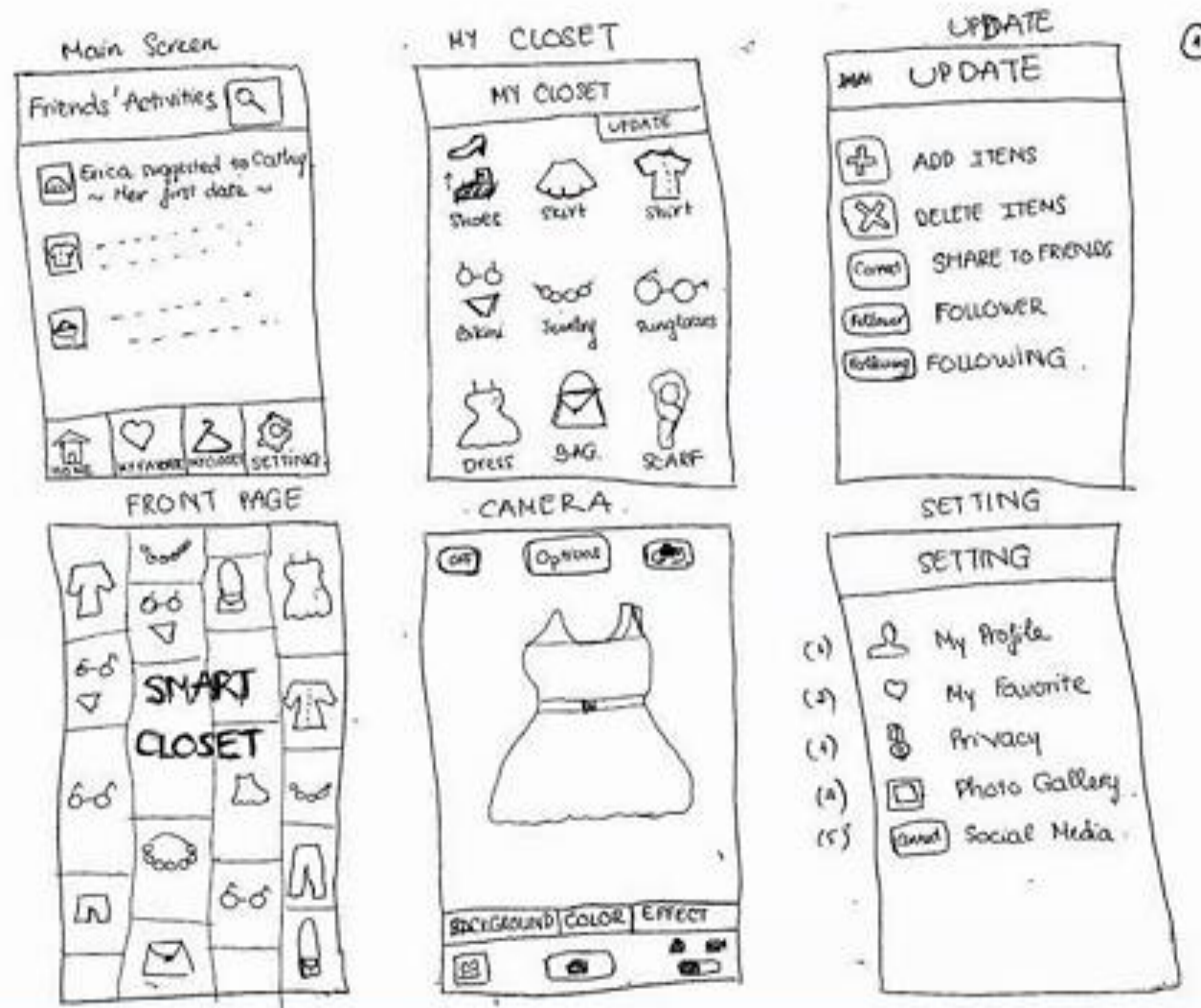


**Keep it
Small**



**Prototypes are not for
Production**

Paper Interfaces



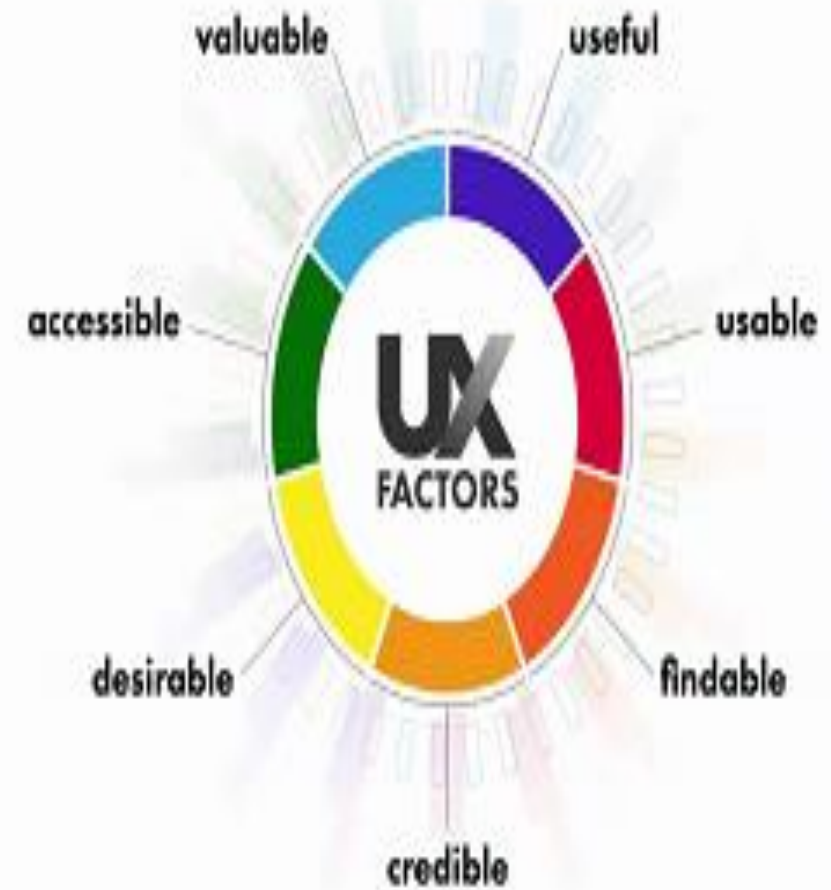
Customer Experience

Company or Brand Focused



User Experience

Product or Service Focused



Storyboards





Ten Usability Heuristics by Jakob Nielsen



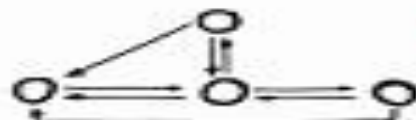
Visibility of system status

Give the users appropriate feedback about what is going on.



Match between system and the real world

Use real-world words, concepts and conventions familiar to the users in a natural and logical order.



User control and freedom

Support undo, redo and exit points to help users leave an unwanted state caused by mistakes.



Error prevention

Prevent problems from occurring; eliminate error-prone conditions or check for them before users commit to the action.



Aesthetic and minimalist design

Don't show irrelevant or rarely needed information since every extra elements diminishes the relevance of the others.



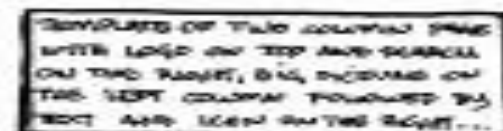
Consistency and standards

Follow platform conventions through consistent words, situations and actions.



Flexibility and efficiency of use

Make the systems efficient for different experience levels through shortcuts, advanced tools and frequent actions.



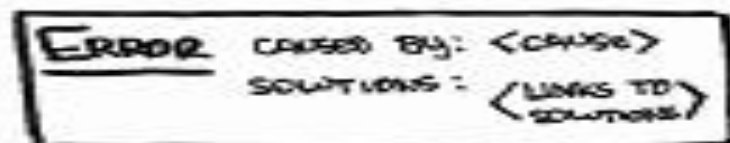
Recognition rather than recall

Make objects, actions, and options visible at the appropriate time to minimize users' memory load and facilitate decisions.



Help and documentation

Make necessary help and documentation easy to find and search, focused



Help users recognize, diagnose, and recover from errors

Express error messages in plain language (no codes) to indicate the problem and suggest solutions.

Why conduct user testing?

- Together, prototyping and testing add huge value to the design process. Not only does user testing help you to remain user-centric; it also makes good business sense. By testing your ideas early and often, you are able to identify design flaws and usability issues before you take the product to market. This has countless benefits for you, the user, and the business!

- **User testing saves time and money:** By catching errors and usability issues early on, you ensure that the product you eventually launch is the most bug-free, user-friendly product it can be. What happens if you skip the testing phase in favor of getting the product developed as soon as possible? You'll spend considerable time and money correcting the product post-launch. Not only is this frustrating—it's also bad for the bottom line.
- **User testing reveals unexpected insights:** No matter how thorough your initial **user research** was, or how convinced you are that you've designed the optimal solution to your user's problem, there are always new insights to be uncovered. **Effective usability testing questions** will highlight issues that you would not have noticed otherwise.

- **User testing improves user satisfaction:** Design Thinking is all about putting the user first. By gathering first-hand user feedback, you can make informed design decisions—improving user satisfaction in the long run. As a designer, prototyping and testing will keep you focused on the user at all times. Of course, satisfied customers are good for business, too!

User Interface Principles

- UI design must take account of the needs, experience and capabilities of the system users.
- Designers should be aware of people's physical and mental limitations (e.g. limited short-term memory) and should recognize that people make mistakes.
- UI design principles underlie interface designs although not all principles are applicable to all designs.

User Interface Principles

<i>Principle</i>	<i>Description</i>
User familiarity	The interface should use terms and concepts which are drawn from the experience of the people who will make most use of the system.
Consistency	The interface should be consistent in that, wherever possible, comparable operations should be activated in the same way.
Minimal surprise	Users should never be surprised by the behaviour of a system.
Recoverability	The interface should include mechanisms to allow users to recover from errors.
User guidance	The interface should provide meaningful feedback when errors occur and provide context-sensitive user help facilities.
User diversity	The interface should provide appropriate interaction facilities for different types of system user.

User Interface Principles

Design principles

- **User familiarity**
 - The interface should be based on user-oriented terms and concepts rather than computer concepts.
 - For example, an office system should use concepts such as letters, documents, folders etc., rather than directories, file identifiers, etc.
- **Consistency**
 - The system should display an appropriate level of consistency. Commands and menus should have the same format, command punctuation should be similar, etc.
- **Minimal surprise**
 - If a command operates in a known way, the user should be able to predict the operation of comparable commands

User Interface Principles

Design principles

- **Recoverability**
 - The system should provide some resilience to user errors and allow the user to recover from errors. This might include an **UNDO** facility, confirmation of destructive actions, 'soft' deletes, etc.
- **User guidance**
 - Some user guidance such as help systems, on-line manuals, etc. should be supplied
- **User diversity**
 - Interaction facilities for different types of user should be supported. For example, some users have seeing difficulties and so larger text should be available

UI Analysis

- **User interface (UI) design** is the study of how users use a particular software application / system or product. **UI design analysis** analyzes users, tasks, content and work environment.
- UI design analysis focuses on the users' **goals**, and the **tasks** they perform within a **system** (the application or product) to achieve those goals. Also content and work environment.
- **User analysis** is the process of studying what type of person might use the application or product in question. For example, are they private individuals or professionals who use it as part of their work?

UI Analysis

- **Task analysis** is the process of studying how users accomplish the tasks they set out to perform using a software system / application or other product. This type of analysis focuses on:
 - What the users' goals are
 - What the users do to achieve those goals
 - The workflow the users follow to accomplish their tasks
 - What the users' level of experience is
 - How the users are affected by their physical environment
- **Content Analysis:** The content that is presented as a part of the interface.
- **Work Environment:** The work environment in which these tasks will be conducted.

UI design

Human Factors In Interface Design

- Limited short-term memory
 - People can instantaneously remember about 7 items of information. If you present more than this, they are more liable to make mistakes.
- People make mistakes
 - When people make mistakes and systems go wrong, inappropriate alarms and messages can increase stress and hence the likelihood of more mistakes.
- People are different
 - People have a wide range of physical capabilities. Designers should not just design for their own capabilities.
- People have different interaction preferences
 - Some like pictures, some like text.

UI components and Responsive design

- Responsive design is a **graphic user interface (GUI) design approach used to create content that adjusts smoothly to various screen sizes**. Designers size elements in relative units (%) and apply media queries, so their designs can automatically adapt to the browser space to ensure content consistency across devices.
- Responsive design is **an approach to web page creation that makes use of flexible layouts, flexible images and cascading style sheet media queries**. The goal of responsive design is to build web pages that detect the visitor's screen size and orientation and change the layout accordingly.

UI and UX process

- **Wire framing**
- Prototyping
- **Testing and iteration**
- Development planning

- Wire framing is a practice used by **UX designers** which allows them to define and plan the information hierarchy of their design for a website, app, or product. This process focuses on how the designer or client wants the user to process information on a site, based on the **user research** already performed by the UX design team.
- When designing for the screen you need to know where all the information is going to go in plain black and white diagrams before building anything with code—**whether that's a developer coding it, or you the designer**. Wire framing is also a great way of getting to know how a user interacts with your interface, through the positioning of buttons and menus on the diagrams.

- Wireframes drawn with paper and a pencil, or at a whiteboard, have the advantage of looking and being very easy to change, which can help tremendously in early conversations about your website or product.
- With the help of paper-prototypes, you can test with end users at every stage of ideation and design. Changes at these stages are much easier—and therefore cheaper—than changes deemed necessary after coding is under way.

The end of Unit - II